Dept . of Computer Science & Engineering

Time : 1.5 hour

Distributed Systems – Class test –I

M.Marks : 30

Q.1.	Part- A (10 Marks) Answer all Questions	M	CO	BT
a)	Define "Scalability" and distinguish between 3 different dimensions of Scalability.	2	1	2
b)	What is the main issue in managing " structured Peer- to Peer " architectures ?	2	2	3
c)	Distinguish between Multi-threading, Single threaded process and Finite State Machine on – parallelism and Blocking System calls.	2	1	3
d)	What is the specific advantage of Virtual Machines? How is it different from para-virtualization ?	2	2	3
e)	What are the different layers that are used in namespace management ?	2	3	3
	Part- B (2 * 10= 20 Marks) Answer any TWO questions			
2. a)	What is " middleware" ? Is it a hardware or software? Should a system be built distributed, not just parallel for implementing modern applications? Justify your answer	5	1	2
b)	In hybrid architectures, elements from centralized and decentralized organizations are combined. For Edge- Cloud architecture based systems analyze Latency, Reliability and Security	5	2	4
3. a)	Using an architecture diagram, bring out the importance of Multithreaded Servers. Analyze the importance of Dispatcher thread and Worker threads.	5	2	3
b)	Distinguish between "Iterative " vs " Recursive" Naming with specific use cases for each of these two approaches.	5	2	3
4. a)	What is the difference between " Process Virtualization", "Hosted Virtual Machine Monitor", and " Native virtual Machine Monitor " ?	5	1	3
b)	Analyze how Vector Clocks overcome the limitations of Lamport's logical clocks ?	5	2	3